

# E rate is affine in gamma frequency

exp041 · 2 June 2026

The trained networks this entry uses are produced once in the shared training hub, exp022 (Training), and reused here rather than retrained.

## Abstract

exp037 varied  $\tau_{\text{GABA}}$  at inference and the per-cell E rate tracked the gamma cycle. Does that survive *re-training* at each  $\tau_{\text{GABA}}$ ? Yes. Across  $\tau_{\text{GABA}} \in \{4.5, 6, 9, 12, 18, 27\}$  ms  $\times$  3 seeds,  $r_E = 1.15 + 0.183 \cdot f_\gamma$  with  $R^2 = 0.994$ . The slope is per-cycle E-cell participation; the intercept is a non-rhythmic baseline; accuracy stays at  $\approx 86\text{--}92\%$  across the sweep. The cycle clock constrains what the network can become.

## Method

**Sweep.** Six  $\tau_{\text{GABA}}$  values  $\{4.5, 6, 9, 12, 18, 27\}$  ms  $\times$  three seeds = 18 networks, trained in the shared hub to the gamma standard (50 epochs on MNIST, Adam at  $4 \times 10^{-4}$ , batch 256, mem-mean readout, no spike budget,  $\Delta t = 0.1$  ms,  $T = 200$  ms); only  $\tau_{\text{GABA}}$  varies.

**Measuring  $f_\gamma$ .** For each cell,  $f_\gamma$  is the parabolic-interpolated peak of the Welch PSD on the per-trial population E trace; the fit (Figure 4) uses per-trial peak medians, which avoid the centroid bias of trial-mean PSD peaks. Parabolic interpolation with peak-bin values  $(y_0, y_1, y_2)$ :

$$f_\gamma = \text{freq}[\text{peak}] + \frac{1}{2} \frac{y_0 - y_2}{y_0 - 2y_1 + y_2} \cdot \Delta f, \quad \Delta f = 5 \text{ Hz}.$$

It is needed because the bare 5 Hz bin quantisation would coarsen  $f_\gamma$  across the six conditions; on a well-isolated peak the interpolation error is  $O((\Delta f)^3)$ .

**The predicted law.** The shape  $r_E = a + p \cdot f_\gamma$  is predicted by cycle dynamics, not curve-fitted. Within one cycle of duration  $1/f_\gamma$ , a fraction  $p$  of E cells emits exactly one spike (those nearest threshold when the I shunt drops); the rest are still recovering, so the cyclic per-cell rate is  $p \cdot f_\gamma$ . At long  $\tau_{\text{GABA}}$  the I conductance never fully decays and the cycle dissolves into a tonic bath, leaving a feedforward baseline  $a$  independent of  $f_\gamma$ :

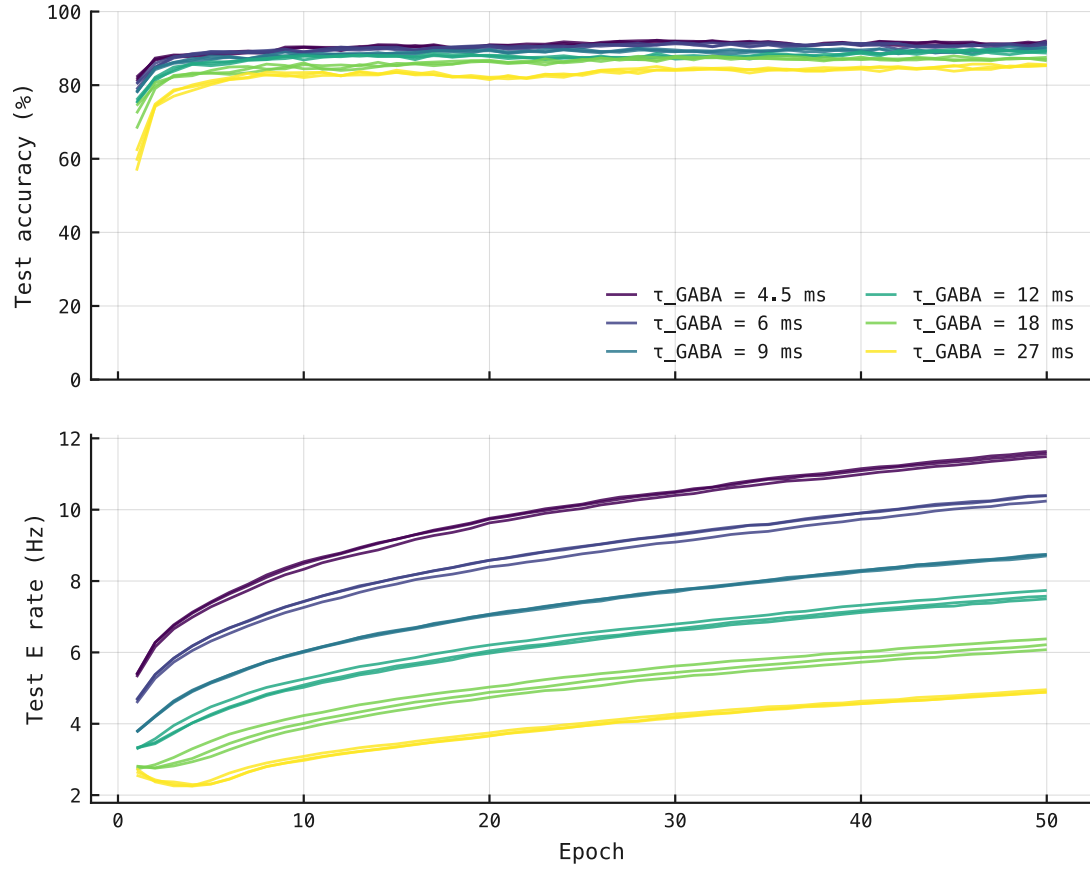
$$r_E = \underbrace{a}_{\text{feedforward baseline}} + \underbrace{p \cdot f_\gamma}_{\text{cyclic contribution}}.$$

We fit this across the 18 cells.

**Convergence.** Accuracy plateaus by  $\approx$  epoch 15 while the E rate keeps climbing through training (Figure 1), so the fit uses the final-epoch rates. Fitting  $r_E = a + p \cdot f_\gamma$  across the 18 cells is tight, and forcing the intercept through zero barely loosens it, so the law is not an artefact of the free intercept:

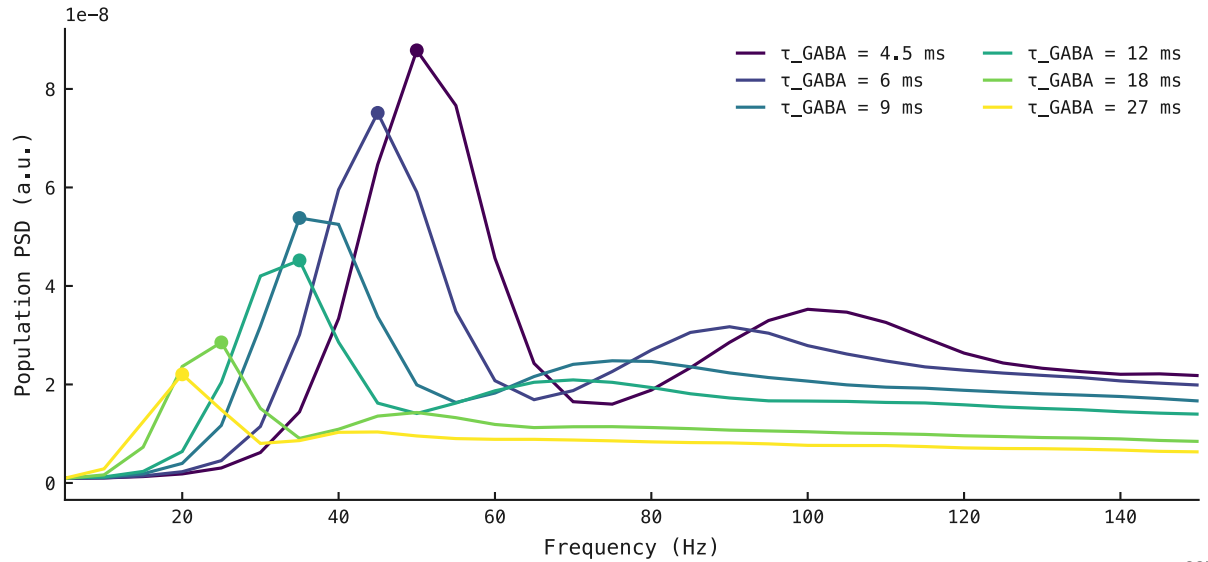
| fit            | $a$ (Hz) | $p$ (Hz/Hz) | $R^2$ |
|----------------|----------|-------------|-------|
| affine         | 1.15     | 0.183       | 0.994 |
| through origin | 0        | 0.213       | 0.966 |

## Results



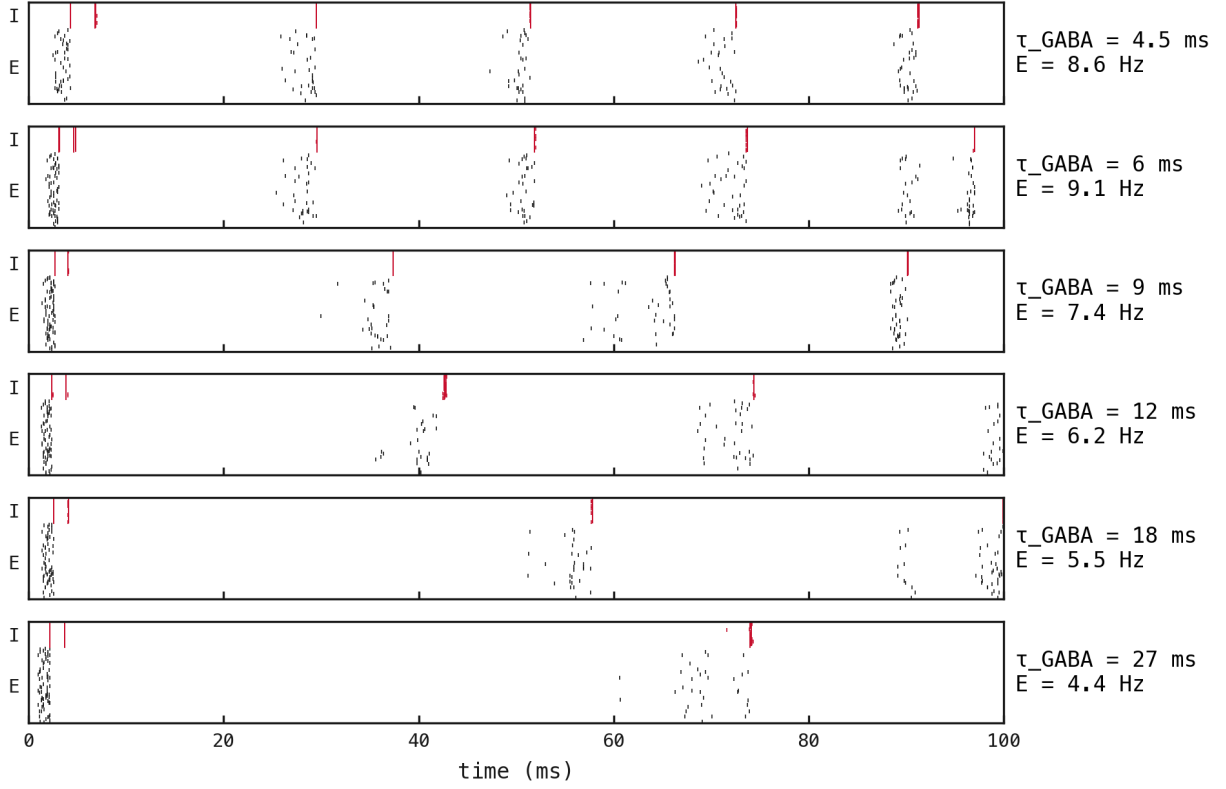
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Figure 1: Per-cell accuracy (top) and E rate (bottom) over training, one line per cell. Accuracy plateaus by  $\approx$  epoch 15; the E rate keeps climbing through the 50 epochs, so the final-epoch rates are the ones fit.



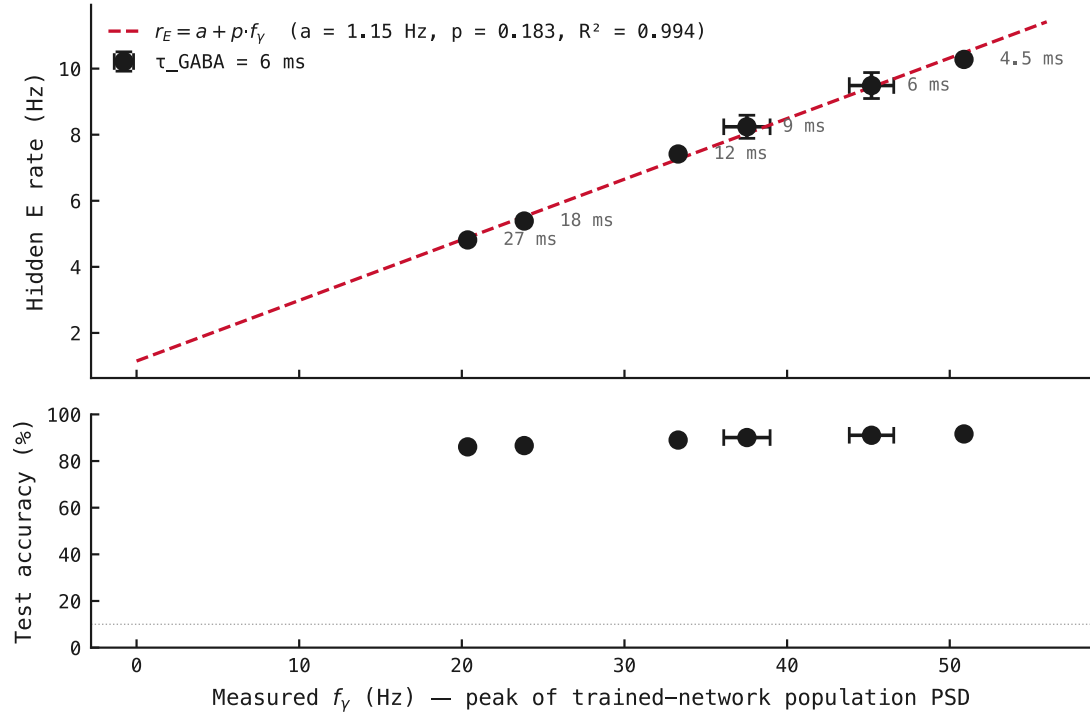
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Figure 2: Trial-mean Welch PSDs by  $\tau_{\text{GABA}}$ ; dots mark the parabolic-interpolated peak. The peak shifts cleanly from  $\approx 20$  Hz at  $\tau_{\text{GABA}} = 27$  ms to  $\approx 51$  Hz at  $\tau_{\text{GABA}} = 4.5$  ms, with no overlap between adjacent conditions.



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Figure 3: One MNIST trial through each network. The cycle period stretches from  $\approx 20$  ms ( $f_\gamma \approx 51$  Hz) at short  $\tau_{\text{GABA}}$  to  $\approx 50$  ms ( $f_\gamma \approx 20$  Hz) at long, the eye and the spectrum agreeing.



r007

Figure 4: The law itself. Top: mean post-training E rate vs  $f_\gamma$ , six clusters  $\times$  three seeds, error bars from seed variance; the affine fit passes through every error bar. Bottom: per-cluster accuracy is flat, so the rate change is not paid in classification.